# PERMANENCE OF SYNTOMIC ALONG A SURJECTIVE FLAT LFP $F$

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted span cites two Morphisms of Spaces lemmas and one Descent lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

*Allowed standard prerequisites (editorial).*

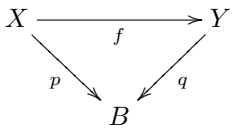
- \'Etale locality of syntomic morphisms. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-syntomic-local. Justifies the reduction to affine $B$ and $Y$.
- Openness of fppf morphisms on points. Stacks Project, Morphisms of Spaces, Lemma spaces-morphisms-lemma-fppf-open. Lets the proof pick $U$ with $U \to Y$ surjective.
- Scheme-side permanence. Stacks Project, Descent, Lemma descent-lemma-syntomic-permanence. Concludes for the scheme diagram.

*Source reference.* Original reference 03ZA: sources/stacks-chow/tags.txt, source lines 5213–5213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 042S: sources/stacks-chow/tags.txt, source lines 5339–5339. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 05B7: sources/stacks-chow/tags.txt, source lines 6875–6875. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $S$ be a scheme. Let*




*be a commutative diagram of morphisms of algebraic spaces over $S$. Assume that*

(1) *$f$ is surjective, flat, and locally of finite presentation,*
(2) *$p$ is syntomic.*

*Then both $q$ and $f$ are syntomic.*

*Proof.* We deduce this from the analogue for schemes. Namely, the problem is étale local on $B$ and $Y$ (Morphisms of Spaces, Lemma 03ZA). Hence we may assume that $B$ and $Y$ are affine schemes. Since $|X| \to |Y|$ is open (Morphisms of Spaces, Lemma 042S), we can choose an affine scheme $U$ and an étale morphism $U \to X$

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such that the composition $U \to Y$ is surjective. In this case the result follows from Descent, Lemma 05B7. $\square$